PHS 2000B Problem Set 9: Missing Data (Solutions)

Due Tuesday, April 29, 2025 at 11:59pm

Part I: A Missing Confounder (12 points)

For a recent research project, you collected data on alcohol abuse as a confounder of the relationship between exposure A and outcome Y. All individuals in the study reported A and Y, but not all individuals reported alcohol abuse. In fact, you suspect that the only people who refused to report alcohol abuse were those who abused alcohol because they felt uncomfortable reporting it. Among alcohol abusers, you suspect that who ends up with a missing value and who ends up with an observed value for alcohol abuse is essentially random (for instance, you can imagine that each alcohol abuser flips a coin to decide whether to complete the alcohol abuse question or leave it unanswered). You have not collected data on any other behavioral variables.

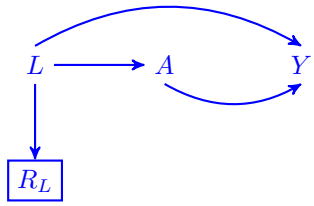
Although it seems that data on alcohol abuse are Missing Not At Random (MNAR) - because the probability of not responding to the question on alcohol abuse is determined by the value of the alcohol abuse variable itself - it's important to think through the implications of missing data within the context of your specific estimand and analytic approach.

With this scenario in mind, please answer the following questions:

1. Draw a DAG corresponding to the missingness scenario described in the prompt. (2 pt)

Answer:

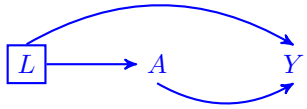
Because there is an unmeasured value of the variable L (alcohol abusers) that predicts the missingness, our missingness pattern is MNAR.



2. If we conducted a complete-case analysis *among non-alcohol abusers,* could we obtain an unbiased estimate of the causal effect of A on Y in this subgroup? Briefly explain, including a DAG that corresponds to this analysis. (3 pt)

Answer:

Yes, we can obtain an unbiased estimate. We do not expect any missingness among non-alcohol abusers, so we can obtain an unbiased estimate in this stratum.

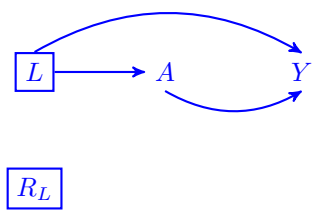


3. Similarly, if we conducted a complete-case analysis *among alcohol abusers,* could we obtain an unbiased estimate of the causal effect of A on Y in this subgroup? Briefly explain, including a DAG that corresponds to this analysis. (3 pt)

Answer:

Yes, we can obtain an unbiased estimate in this case. Among alcohol abusers, we are told that missingness is essentially random (i.e., like tossing a coin). As a result, people in the alcohol abuse stratum are a random subsample of all alcohol abusers. Even though we are missing observations in this stratum, we can obtain an unbiased estimate because missingness is completely random (MCAR) among alcohol abusers.

In the subset of alcohol abusers, missingness is completely random (i.e., there are no arrows into the missingness indicator):



4. If we assume no effect modification by alcohol abuse, could we obtain an unbiased estimate of the average (i.e., marginal) causal effect of A on Y using a complete case analysis? Briefly explain your answer. (2 pts)

Answer:

If we can assume that there is no effect modification by alcohol abuse, then the marginal average causal effect would be unbiased. The logic is that a random subsample of the alcohol abusers were excluded from the study. That is, under the assumption of no effect modification (i.e., the effect is the same among alcohol abusers and non-abusers), we would expect our estimate from a complete-case analysis to provide an unbiased estimate of the average causal effect.

We must assume no effect modification (on the scale of interest for the analysis) because the marginal estimate of the average causal effect obtained from a complete case analysis will reflect the distribution of alcohol abusers in the selected population with missing data, which differs from the distribution in the non-selected population. Thus, if there were in fact effect modification by alcohol abuse on the scale of interest, then the effect estimates in the selected population would not necessarily reflect the average causal effect that would have been estimated had everyone been selected.

OPTIONAL ADDITIONAL EXPLANATION: We can use the first DAG shown above to formalize our conclusions. The distribution that we observe is conditional on having non-missing values:

$$f(Y, A, L|R_L = 1)$$

By applying Bayes theorem (twice), we can factorize this distribution as follows:

$$f(Y, A, L|R_L = 1) = f(Y|A, L, R_L = 1) \times f(A, L|R_L = 1) = f(Y|A, L, R_L = 1) \times f(A|L, R_L = 1) \times f(L|R_L = 1)$$

The first term in the factorized distribution, $f(Y|A, L, R_L = 1)$, represents the distribution of Y conditional on A and L and having observed data on L, $R_L = 1$. Can we rewrite this conditional distribution without conditioning on $R_L = 1$? Yes! Because we can see in the first DAG from question 1 that, if we condition on (put boxes around) L and A, then there will be no open path between Y and R_L. Therefore Y is independent of R_L conditional on A and L and we can rewrite the first term of the factorized distribution as follows:

$$f(Y|A, L, R_L = 1) = f(Y|A, L)$$

The above equality implies that we can use a complete case analysis conditional on A and L in the data with missing values to unbiasedly estimate the conditional effect of A on Y (conditional on L) in the target

population. If we additionally assume no effect modification by L, then we can also identify the average (marginal) causal effect because:

$$f(Y|A, L = 1) = f(Y|A, L = 0) = f(Y|A, L)$$

5. Considering the missingness mechanism in this scenario, would using multiple imputation yield an unbiased estimate of the average causal effect of A on Y? Briefly explain your answer. (2 pts)

Answer:

No, we would not expect multiple imputation to yield an unbiased estimate. Multiple imputation generally requires that the data are MAR (or MCAR) to produce unbiased effects. In this case, the data are MNAR so multiple imputation would not be expected to produce an unbiased effect.

OPTIONAL ADDITIONAL EXPLANATION: More formally, you could also note that multiple imputation aims to reconstruct the joint distribution had no one had missing data, i.e., $f(Y, A, L)$. To reconstruct this distribution, we would need to be able to rewrite every piece of the factorized distribution shown above without conditioning on $R_L = 1$. However, we can see that $f(L|R_L = 1) \neq f(L)$ because of the direct arrow from L to R_L. Therefore, we cannot use multiple imputation to reconstruct the joint distribution had no one had missing data.

Part II: Multiple Imputation (28 points)

Electronic bullying has a number of serious consequences for adolescents. We'll be using the 2015 Youth Risk Behavior Survey (YRBS) to investigate the association between electronic bullying and suicidal ideation (serious consideration of suicide), adjusting for age, sex, number of hours per day spent on a computer, and having been in a physical fight.

The variables which we will need for this analysis are described below:

- *Q25*: During the past 12 months, have you ever been electronically bullied? (1=Yes, 2=No)
- *Q27*: During the past 12 months, did you ever seriously consider attempting suicide? (1=Yes, 2=No)
- *Q1*: How old are you (in years)? (1= $\leq$ 12, 2=13, 3=14, 4=15, 5=16, 6=17, 7= $\geq$ 18)
- *Q2*: What is your sex? (1=Female, 2=Male)
- *Q82*: On an average school day, how many hours do you play video or computer games or use a computer for something that is not school work? (1=0 hours, 2= <1 hour, 3=1 hour, 4=2 hours, 5=3 hours, 6=4 hours 7=5+ hours)
- *QN18*: Student was in a physical fight one or more times during past 12 months (1=Yes, 2=No)

The full data dictionary for the 2015 YRBS data can be found on Canvas.

Note that the YRBS survey asked about sex in a way that does not reflect the richness and diversity of the world in which we live (e.g., the survey did not explicitly attempt to include transsexual individuals and did not ask separately about gender identity, among other shortcomings). In this problem set, the original YRBS data and questions have been retained to remind us that these shortcomings still exist in many of the datasets we use. To pretend otherwise would be to erase an injustice that has been present for decades. We hope that highlighting the injustice serves as a reminder that we can strive to do better when we collect our own primary data.

```r
# Load YRBS data
if (!require("haven")) install.packages("haven")
library("haven")

yrbs2015 <- data.frame(read_sas("yrbs2015.sas7bdat"))

# Perform data cleaning
# Turn blanks into "NA"
yrbs2015[yrbs2015 == ""] <- NA
# Convert some fields from character to numeric
convert <- colnames(yrbs2015[1:99])
yrbs2015[,convert] <-
    lapply(yrbs2015[,convert,drop=FALSE],as.numeric)
```

1. Describe the pattern of missing data in the six variables listed above. Use at least one figure to support your statements and be sure to describe (i) the magnitude of the missingness problem, (ii) any obvious correlation between missingness in multiple variables, and (iii) whether or not the missingness is monotone. (4 pts)

Answer:

All of the variables have some missingness.

i) The magnitude of missing data is moderate overall. Question N18 (which asks whether or not the adolescent got into a fight) has the most missing data.

ii) There appears to be some correlation between missingness in the exposure (Q25) and outcome of interest (Q27), but there are no other obvious patterns or correlations.

iii) The data is not monotone missing (i.e., there is no monotonic ordering with regards to the amount of missing data in each variable). For example, if the data were monotone missing, then individuals

would only be missing data on Q1 (i.e., the variable with the least amount of missing values) if they were also missing all other variables in the dataset.

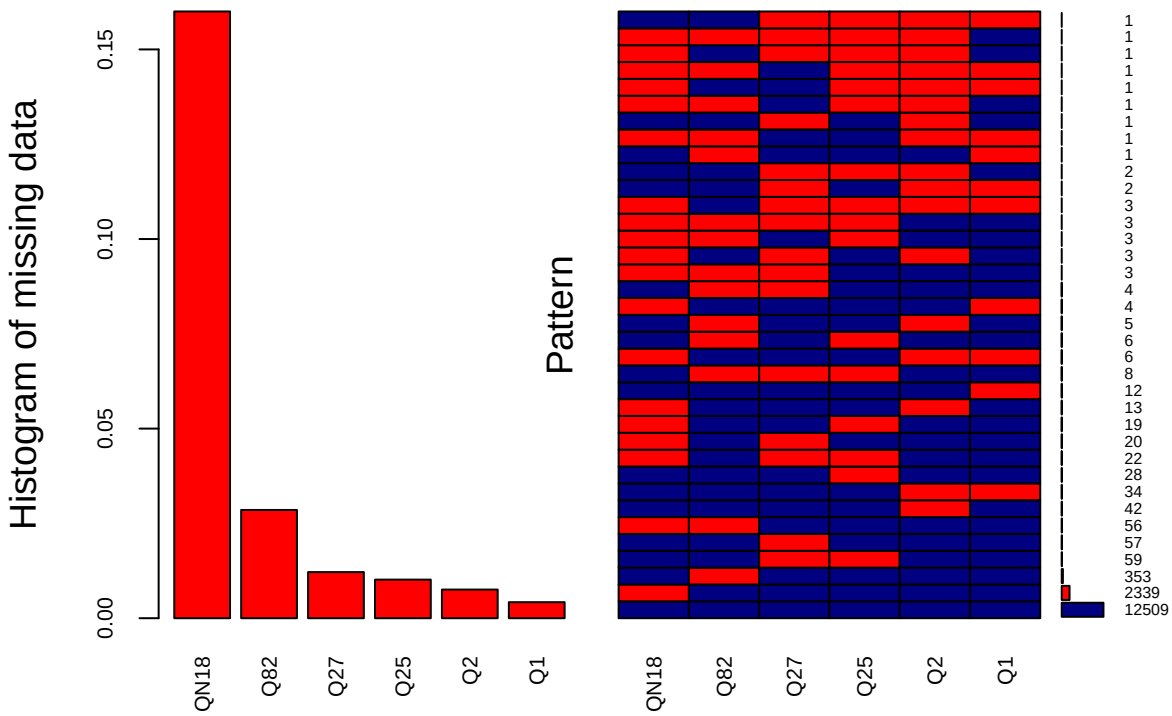
```
# Subset to the 6 variables of interest
yrbs2015.2 <- subset(yrbs2015, select = c(Q25, Q27, Q1, Q2, Q82, QN18))

# Create a function to calculate the proportion of missing data
pMiss <- function(x){sum(is.na(x)) / length(x)*100}

# Apply the missing data function to the subsetted data
apply(yrbs2015.2, 2, pMiss)
```

```
##    Q25    Q27     Q1     Q2    Q82   QN18
##  1.018  1.216  0.422  0.755  2.855 16.001
```

```
# Plot the pattern of missing data
if (!require("VIM")) install.packages("VIM")
library("VIM")
aggr_plot <- aggr(yrbs2015.2, col=c('navyblue','red'), numbers = TRUE, sortVars = TRUE,
                  labels = names(data), cex.axis = 0.7, gap = 1, prop = c(TRUE, FALSE),
                  ylab = c("Histogram of missing data", "Pattern"), cex.numbers = 0.5)
```




```
##
##  Variables sorted by number of missings:
##  Variable   Count
##      QN18 0.16001
##       Q82 0.02855
##       Q27 0.01216
##       Q25 0.01018
##        Q2 0.00755
##        Q1 0.00422
```

2. You hypothesize that participants' grades in school and weapon carrying may be related to missingness.

Do the two variables appear to be predictors of whether the student got into a fight in the last 12 months (the variable with the most missingness)? Provide some support for your response in the form of a figure, frequency table, or regression model results. (4 pts)

Answer:

Answers should be backed up with a figure, frequency table, or regression model results. Both variables (grades in school and weapon carrying) have significant associations with whether the student got into a fight in the last 12 months. Both predictors were significantly associated with getting into a fight across all levels ($p<0.05$). This is seen via regression output and frequency tables. Therefore, these variables appear to be predictors of whether the student got into a fight in the last 12 months.

Note: +1 for including relevant output, +1 point for justification with output (+0.5 for each variable),+1 point for stating grades in school is a predictor of getting a fight in last 12 months, +1 point for stating carrying a weapon is a predictor of getting a fight in last 12 months

An example using `regression` is shown below.

```r
# Load table1 package
library("table1")

# Relabel QN18 for the table output
yrbs2015$Fight <-
  factor(yrbs2015$QN18,
         levels=c(1,0),
         labels=c("Fight in last 12 months",
                  "No fight in last 12 months"))

# GLM for Q89
model_q89<-glm(Fight ~ as.factor(Q89), family=binomial(link="logit"), data=yrbs2015)
summary(model_q89)
```

```
##
## Call:
## glm(formula = Fight ~ as.factor(Q89), family = binomial(link = "logit"),
##     data = yrbs2015)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -2.66e+01   1.57e+04       0        1
## as.factor(Q89)2 -8.15e-26   1.96e+04       0        1
## as.factor(Q89)3 -8.79e-26   2.02e+04       0        1
## as.factor(Q89)4 -1.34e-12   3.00e+04       0        1
## as.factor(Q89)5 -1.04e-25   3.78e+04       0        1
## as.factor(Q89)6 -1.01e-25   6.22e+04       0        1
## as.factor(Q89)7 -9.70e-26   3.74e+04       0        1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 0.0000e+00  on 2658  degrees of freedom
## Residual deviance: 1.5426e-08  on 2652  degrees of freedom
##   (12965 observations deleted due to missingness)
## AIC: 14
##
## Number of Fisher Scoring iterations: 25
```

```r
# GLM for Q13
model_q13<-glm(Fight ~ as.factor(Q13), family=binomial(link="logit"), data=yrbs2015)
summary(model_q13)
```

```
##
## Call:
## glm(formula = Fight ~ as.factor(Q13), family = binomial(link = "logit"),
##     data = yrbs2015)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)     -2.66e+01   8.46e+03       0        1
## as.factor(Q13)2 -1.22e-13   2.85e+04       0        1
## as.factor(Q13)3 -1.22e-13   2.63e+04       0        1
## as.factor(Q13)4 -1.22e-13   4.00e+04       0        1
## as.factor(Q13)5 -1.22e-13   1.78e+04       0        1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 0.0000e+00  on 2745  degrees of freedom
## Residual deviance: 1.5931e-08  on 2741  degrees of freedom
##   (12878 observations deleted due to missingness)
## AIC: 10
##
## Number of Fisher Scoring iterations: 25
```

3. Run a complete-case logistic regression model and provide an estimate of the odds ratio (and its 95% CI) for suicidal ideation comparing those who did vs. did not experience electronic bullying, adjusting for age, sex, number of hours per day spent on a computer, and having been in a physical fight. Interpret the odds ratio. *Note: For the purposes of this analysis, treat age and number of hours spent on a computer as continuous variables and ignore any product terms or non-linearities.* (2 pts)

Answer:

Electronic bullying is associated with 3.66 times the odds (95% CI: 3.27, 4.10) of suicidal consideration, conditional on age, sex, number of hours per day spent on a computer, and having been in a physical fight.

Note: -0.5 if did not include conditional statement.

```r
# First, convert dichotomous variables to 0/1 instead of 1/2 for interpretability and modeling
yrbs2015$Q25[yrbs2015$Q25 == 2] <- 0
yrbs2015$Q27[yrbs2015$Q27 == 2] <- 0
yrbs2015$Q2[yrbs2015$Q2 == 2] <- 0
yrbs2015$QN18[yrbs2015$QN18 == 2] <- 0

# Fit the logistic model
CCmodel <- glm(Q27 ~ Q25 + Q1 + Q2 + Q82 + QN18,
               family = binomial(link = "logit"),
               data = yrbs2015)
summary(CCmodel)
```

```
##
## Call:
## glm(formula = Q27 ~ Q25 + Q1 + Q2 + Q82 + QN18, family = binomial(link = "logit"),
##     data = yrbs2015)
##
```

```
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -2.8393     0.1293  -21.96  < 2e-16 ***
## Q25           1.2974     0.0576   22.53  < 2e-16 ***
## Q1           -0.0031     0.0204   -0.15     0.88
## Q2            0.8413     0.0528   15.94  < 2e-16 ***
## Q82           0.0898     0.0115    7.84  4.4e-15 ***
## QN18          0.8338     0.0559   14.91  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 11788  on 12508  degrees of freedom
## Residual deviance: 10561  on 12503  degrees of freedom
##   (3115 observations deleted due to missingness)
## AIC: 10573
##
## Number of Fisher Scoring iterations: 5
```

```r
# Print the odds ratio for the effect of electronic bullying on suicidal ideation, along with its 95% C
exp(cbind(OR = coef(CCmodel), confint(CCmodel)))
```

```
##                 OR  2.5 % 97.5 %
## (Intercept) 0.0585 0.0453 0.0752
## Q25         3.6599 3.2688 4.0968
## Q1          0.9969 0.9579 1.0375
## Q2          2.3193 2.0922 2.5730
## Q82         1.0939 1.0697 1.1188
## QN18        2.3021 2.0627 2.5684
```

4. Use the `mice` package in R to perform multiple imputation. Create $m = 5$ imputed datasets. At a minimum, your imputation model should include (i) the two variables you chose in Question 2 and (ii) all study variables. Briefly describe the imputation algorithm you used and any modeling assumption(s) that were made (5 sentences maximum). (3 pts)

Answer:

Multiple imputation by chained equations is a sequential approach in which the distribution of each variable with missingness is modeled by all other variables (i.e., a fully conditional specification for each missing variable). For each imputed dataset, the algorithm uses simple means to initialize roughly reasonable values for the unknown information (i.e., the missing values as well as the parameters for each imputation model). Then, it runs a Gibbs sampler in which each iterate (1) refits the imputation model to the currently imputed data, and (2) draws new values for the missing values from their respective posterior predictive distributions. For each of $m = 5$ datasets, the sampler iterates until convergence, at which point we have one of our imputed datasets.

To impute missing values, we used the default, predictive mean matching. With predictive mean matching, each missing value is drawn from a "donor pool" of close matches based on the other variables in the model. This approach does not make strong parametric assumptions - the imputed values are drawn from the observed distribution. Specifically, predicted values of the target missing variables are calculated according to a parametric regression model, and then a small pool of donors (typically 3-10) is formed. Donors are complete cases who have predicted values close to the predicted value for the observation which is to be imputed. Therefore, when using predictive mean matching **we essentially assume that the distribution of the missing observations is the same as the observed distribution among the candidate donors.** We don't need to specify an explicit model for missingness, so the method is more robust to model misspecification than other approaches. Unlike other multiple imputation approaches (e.g., AMELIA II),

MICE does not require us to assume that the joint distribution is multivariate normal.

```r
# First, restrict to variables which will be used in the imputation model
to_keep <- c("Q25", "Q27", "Q1", "Q2", "Q82", "QN18", "Q89", "Q13") # UPDATE LIST OF VARIABLES
d <- yrbs2015[ , to_keep ]

# Call the MICE package
if (!require("mice")) install.packages("mice")
library(mice)

# Implement MICE with m=5 imputed datasets
imps <- mice(d, m=5)
```

```
##
##  iter imp variable
##   1   1  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   1   2  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   1   3  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   1   4  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   1   5  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   2   1  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   2   2  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   2   3  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   2   4  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   2   5  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   3   1  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   3   2  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   3   3  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   3   4  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   3   5  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   4   1  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   4   2  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   4   3  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   4   4  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   4   5  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   5   1  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   5   2  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   5   3  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   5   4  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
##   5   5  Q25  Q27  Q1  Q2  Q82  QN18  Q89  Q13
```

```r
# Check the predictor matrix used
imps$pred
```

```
##      Q25 Q27 Q1 Q2 Q82 QN18 Q89 Q13
## Q25    0   1  1  1   1    1   1   1
## Q27    1   0  1  1   1    1   1   1
## Q1     1   1  0  1   1    1   1   1
## Q2     1   1  1  0   1    1   1   1
## Q82    1   1  1  1   0    1   1   1
## QN18   1   1  1  1   1    0   1   1
## Q89    1   1  1  1   1    1   0   1
## Q13    1   1  1  1   1    1   1   0
```

```r
# Check the default imputation method
imps$method
```

```
##   Q25   Q27    Q1    Q2   Q82  QN18   Q89   Q13
## "pmm" "pmm" "pmm" "pmm" "pmm" "pmm" "pmm" "pmm"
```

5. For the variable with the most missingness (i.e., whether the student got into a fight in the last 12 months), provide a table or figure describing this variable's distribution in both the observed data and in *one* of the imputed datasets you created in Question 5. Consider both (i) the marginal distribution of the variable getting into a fight in the last 12 months and (ii) the conditional distribution of the variable getting into a fight in the last 12 months, conditional on one other variable which you hypothesize may be predictive of missingness. Why might you see differences between these distributions (i.e., when comparing the marginal and conditional distributions in the observed data to those same distributions in one of the imputed datasets), even if you modeled the missingness correctly? (4 pts)

Answer:

Tables/figures will vary.

Plausible explanations for the differences in the marginal and conditional distributions in the observed vs. imputed datasets include: (1) The distributions in the observed vs. imputed datasets could differ due to small sample sizes (stochastic or random differences) (2) The two distributions in the observed vs. imputed datasets could differ because there are strong predictors of the missing values included in the imputation model (i.e., some respondents with missing data were more or less likely to get into a fight than others, and we were able to predict these missing values with the other covariates included in the imputation model). For the marginal distributions, if missingness is MAR, then we would expect some change in the marginal distribution on average if we included predictors of missingness. For the conditional distribution conditional on a single variable, these may also differ if there are other variables (which we have not conditioned on) that were important in the imputation procedure.

Note: +1 point for marginal distribution, + 1 point for conditional distirbution, +2 point for justification

```r
# Extract the first imputed dataset
mouse <- mice::complete(imps, 1)

# Concatenate observed data with the first imputed dataset (for ease of analysis)
temp <- rbind(d, mouse)
temp$dataset <- c(rep( "Original", nrow(d) ),
                  rep( "Imputation m=1", nrow(mouse) ) )

# Display the marginal distributions in the observed vs. imputed data
library(dplyr)
library(tidyr)

temp |>
  group_by(dataset) |>
  filter( !is.na(QN18) ) |>
  summarize( prop.yes = mean(QN18) )
```

```
## # A tibble: 2 x 2
##   dataset        prop.yes
##   <chr>             <dbl>
## 1 Imputation m=1    0.224
## 2 Original          0.220
```

```r
# Display the conditional distributions in the observed vs. imputed data
# conditioning on a variable hypothesized to be predictive of missingness
temp |>
  group_by(dataset, Q13) |>
  filter( !is.na(QN18) & !is.na(Q13) ) |>
  summarize( prop.yes = mean(QN18) )
```

```
## # A tibble: 10 x 3
## # Groups:   dataset [2]
##    dataset          Q13 prop.yes
##    <chr>          <dbl>    <dbl>
##  1 Imputation m=1     1    0.170
##  2 Imputation m=1     2    0.389
##  3 Imputation m=1     3    0.429
##  4 Imputation m=1     4    0.504
##  5 Imputation m=1     5    0.526
##  6 Original           1    0.165
##  7 Original           2    0.420
##  8 Original           3    0.431
##  9 Original           4    0.488
## 10 Original           5    0.507
```

```
temp |>
  group_by(dataset, Q89) |>
  filter( !is.na(QN18) & !is.na(Q89) ) |>
  summarize( prop.yes = mean(QN18) )
```

```
## # A tibble: 14 x 3
## # Groups:   dataset [2]
##    dataset          Q89 prop.yes
##    <chr>          <dbl>    <dbl>
##  1 Imputation m=1     1    0.149
##  2 Imputation m=1     2    0.200
##  3 Imputation m=1     3    0.306
##  4 Imputation m=1     4    0.406
##  5 Imputation m=1     5    0.515
##  6 Imputation m=1     6    0.450
##  7 Imputation m=1     7    0.315
##  8 Original           1    0.136
##  9 Original           2    0.191
## 10 Original           3    0.297
## 11 Original           4    0.401
## 12 Original           5    0.517
## 13 Original           6    0.412
## 14 Original           7    0.264
```

6. Obtain a pooled estimate across the 5 imputations for the odds ratio of suicidal ideation comparing those who did vs. did not experience electronic bullying, adjusting for age, sex, number of hours per day spent on a computer, and having been in a physical fight. Briefly comment on whether this pooled estimate is similar to or different from the estimate you obtained in Question 4. (3 pts)

Answer:

The estimated odds ratio is 3.90, which is a little larger than the odds ratio computed in Question 4. The complete case analysis we implemented in Question 4 would generally be a valid estimate of the conditional causal effect if the data were MAR conditional on the variables included in the model. The missingness must also not depend on the outcome and exposure non-independently (NOTE: this is a smaller nuance that we did not discuss in depth. Thorough understanding of this information would not be required on the exam).

Multiple imputation using MICE assumes the data are MAR conditional on the variables included in the imputation model (note that we included some additional variables that were not included in the outcome regression). With predictive mean matching, we also assume the distribution of the missing observation is the same as the observed distribution among the candidate donors.

These differences in assumptions, along with sampling error, could account for the slight difference in point estimates between the two methods.

Note: +1 point for pooled estimate, +0.5 point for comparison to Question 4 in magnitude, +1.5 point for justification

```
# Fit analytic model within each of the m imputed datasets
impmods <- with( imps, glm(Q27 ~ Q25 + Q1 + Q2 + Q82 + QN18,
                           family = binomial(link = "logit") ) )
# Pool across 5 datasets
pooled <- pool(impmods)

# Compute the pooled OR
str(pooled)   # This command helps us to see the structure of the object "pooled"
```

```
## Classes 'mipo' and 'data.frame': 0 obs. of  4 variables:
##  $ call   : language pool(object = impmods)
##  $ m      : int 5
##  $ pooled :'data.frame': 6 obs. of  11 variables:
##   ..$ term    : Factor w/ 6 levels "(Intercept)",..: 1 2 3 4 5 6
##   ..$ m       : int  5 5 5 5 5 5
##   ..$ estimate: num  -2.88891 1.36289 0.00959 0.80306 0.09014 ...
##   ..$ ubar    : num  0.012669 0.0026 0.000319 0.002187 0.000103 ...
##   ..$ b       : num  0.00018127 0.00008447 0.00000064 0.00006923 0.00000157 ...
##   ..$ t       : num  0.012887 0.002701 0.00032 0.00227 0.000105 ...
##   ..$ dfcom   : num  15618 15618 15618 15618 15618 ...
##   ..$ df      : num  7333 2389 15236 2492 6819 ...
##   ..$ riv     : num  0.01717 0.03899 0.00241 0.03798 0.01838 ...
##   ..$ lambda  : num  0.0169 0.0375 0.0024 0.0366 0.018 ...
##   ..$ fmi     : num  0.01715 0.03833 0.00253 0.03737 0.01834 ...
##  $ glanced:'data.frame': 5 obs. of  8 variables:
##   ..$ null.deviance: num  14862 14859 14847 14853 14853
##   ..$ df.null      : int  15623 15623 15623 15623 15623
##   ..$ logLik       : num  -6605 -6618 -6626 -6631 -6618
##   ..$ AIC          : num  13221 13247 13264 13275 13248
##   ..$ BIC          : num  13267 13293 13310 13321 13294
##   ..$ deviance     : num  13209 13235 13252 13263 13236
##   ..$ df.residual  : num  15618 15618 15618 15618 15618
##   ..$ nobs         : int  15624 15624 15624 15624 15624
```

```
exp(pooled$pooled$estimate)
```

```
## [1] 0.0556 3.9075 1.0096 2.2324 1.0943 2.3330
```

7. Report estimates of (i) the relative increase in variance due to non-response (r) and (ii) the fraction of missing information ($\hat{\lambda}$) for the log-odds ratio of suicidal ideation comparing those who did vs. did not experience electronic bullying. Comment on the magnitude of these values and what this tells you about inferential uncertainty in your $\hat{\beta}$. (4 pts)

Answer:

(r) and ($\hat{\lambda}$) can be useful diagnostics for assessing how missing data contributes to inferential uncertainty about our estimates.

In this multiple imputed dataset $r = 0.0306$, which indicates that the relative increase in variance due to nonresponse for the log odds ratio of suicidal ideation comparing those who did vs. did not experience electronic bullying is 3.1%. $\lambda = 0.0296$ which indicates that the fraction of missing information is 3.0%. Both of these values indicates relatively little inferential uncertainty.

Note: +1 for estimate r, +1 for $\hat{\lambda}$, +2 for justification

```
pooled$pooled$r
```

```
## [1] 0.01717 0.03899 0.00241 0.03798 0.01838 0.31736
```

```
pooled$pooled$lambda
```

```
## [1] 0.0169 0.0375 0.0024 0.0366 0.0180 0.2409
```

8. To understand how pooled estimates are calculated, we will now implement Rubin's rules by hand. For simplicity of illustration, we'll assume that we used OLS instead of logistic regression to answer the prior question (although Rubin's rules may also be applied to logistic regression) and that we created only two imputed datasets. We obtain the following results (via the code below) after running the OLS model in the two imputed datasets. Provide a *pooled estimate* for (i) the beta coefficient associated with electronic bullying and (ii) the standard error of this beta coefficient. Show your work. (4 pts)

```
if (!require("mice")) install.packages("mice")
library("mice")

yrbs2015$Q25[yrbs2015$Q25 == 2] <- 0
yrbs2015$Q27[yrbs2015$Q27 == 2] <- 0
yrbs2015$Q2[yrbs2015$Q2 == 2] <- 0
yrbs2015$QN18[yrbs2015$QN18 == 2] <- 0

yrbs2015.2 <- subset(yrbs2015, select = c(Q25, Q27, Q1, Q2, Q82, QN18))

impdata1 <- mice(yrbs2015.2, m=2,
                 meth=c("logreg", "logreg", "pmm", "logreg", "pmm", "logreg"),
                 seed=500)
```

```
##
##  iter imp variable
##   1   1  Q25  Q27  Q1  Q2  Q82  QN18
##   1   2  Q25  Q27  Q1  Q2  Q82  QN18
##   2   1  Q25  Q27  Q1  Q2  Q82  QN18
##   2   2  Q25  Q27  Q1  Q2  Q82  QN18
##   3   1  Q25  Q27  Q1  Q2  Q82  QN18
##   3   2  Q25  Q27  Q1  Q2  Q82  QN18
##   4   1  Q25  Q27  Q1  Q2  Q82  QN18
##   4   2  Q25  Q27  Q1  Q2  Q82  QN18
##   5   1  Q25  Q27  Q1  Q2  Q82  QN18
##   5   2  Q25  Q27  Q1  Q2  Q82  QN18
```

```
MICEresults1 <- with(impdata1, lm(Q27 ~ Q25 + Q1 + Q2 + Q82 + QN18))
```

```
summary(MICEresults1)
```

```
## # A tibble: 12 x 7
##    term        estimate std.error statistic   p.value  nobs df.residual
##    <chr>          <dbl>     <dbl>     <dbl>     <dbl> <int>       <dbl>
##  1 (Intercept) 0.00780    0.0144     0.543  5.87e-  1 15624       15618
##  2 Q25         0.261      0.00844   31.0    1.27e-204 15624       15618
##  3 Q1          0.00118    0.00237    0.499  6.18e-  1 15624       15618
##  4 Q2          0.105      0.00596   17.6    1.45e- 68 15624       15618
##  5 Q82         0.0125     0.00135    9.31   1.51e- 20 15624       15618
##  6 QN18        0.123      0.00715   17.2    1.15e- 65 15624       15618
##  7 (Intercept) 0.000865   0.0143     0.0606 9.52e-  1 15624       15618
```

```
##  8 Q25          0.257      0.00843   30.5    7.51e-199 15624       15618
##  9 Q1           0.00205    0.00236    0.870  3.84e-  1 15624       15618
## 10 Q2           0.107      0.00594   18.1    1.90e- 72 15624       15618
## 11 Q82          0.0121     0.00135    9.02   2.13e- 19 15624       15618
## 12 QN18         0.137      0.00714   19.2    5.56e- 81 15624       15618
```

Answer:

Beta Estimate:

$$\bar{\beta} = \frac{1}{m}\sum_{k=1}^{m}\hat{\beta}^{(k)}$$

$$= \frac{1}{2}(0.261 + 0.257) = 0.259$$

Standard Error Estimate:

$$T = \bar{V_\beta} + (1 + \frac{1}{m})B$$

$$\bar{V_\beta} = \frac{1}{m}\sum_{k=1}^{m}\hat{V}_\beta^{(k)}$$

$$\bar{V_\beta} = \frac{1}{2}(0.00844^2 + 0.00843^2) = 0.0000711$$

$$B = \frac{1}{m-1}\sum_{k=1}^{m}(\hat{\beta}^{(k)} - \bar{\beta})^2$$

$$B = \frac{1}{1}[(0.261 - 0.259)^2 + (0.257 - 0.259)^2] = 0.000008$$

$$T = 0.0000711 + (1 + \frac{1}{2})0.000008 = 0.0000831$$

$$S\hat{E}_\beta = \sqrt{0.0000831} = 0.00912$$

Note: +2 for each estimate